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Q.1) Double Number pyramid
for n = n

```
import java.util.*;
Class Demo {
Public static void main (String args[]) {
Scanner scan = new Scanner (System.in);
int n = scan.nextInt();
for (int i=1 ; i<=n ; i++) {
    for (int j = 1 ; j<=n-i ; j++) {
        sop (" ");
    }
    for (int j=1 ; j<=(2*i)-i ; j++) {
        sop(j+" ");
    }
    Sopln();
}
}
for (int i = n-1 ; i>=1 ; i--) {
    for (int j = 1 ; j<=n-i ; j++) {
        sop(" ");
    }
    for (int j = 1 ; j<=(2*i)-1 ; j++) {
        sop (j+" ");
    }
    Sopln();
}
}
```

Output

```
1
12
123
1234
```

4x4

nxn

```
123
12
1
```

~~4-i~~

n - i = space

4 - i = spa = 3

for 1 ≤ 4

j = i = 1

j ≤ 4

i = 1 j = 1 space

j ≤ 2

i = 1 j = 2 → space

2 ≤ 4

i ≤ 1 j = 3 → space

3 ≤ 4

i = 1 j = 4 space = 1

4 ≤ 4

~~i = 1 j = 5~~

5 ≤ 5 ✗

n - i = space

4 - 1 = 3 space

(2 × i) - i

2 × 1 = 1

2 - 1 = 1

i = 2 2 ≤ 4

i = 2 j = 1 space

j ≤ 4

i = 2 j = 2 space

2 ≤ 4

i = 2 j = 3 → 1

3 ≤ 4

i = 2 j = 4 → 2

4 ≤ 4

~~i = 2 j = 5~~

5 ≤ 5 ✗

n - i = space

4 - 2 = 2 spaces

(2 × i) - i

2 × 2 = 2

4 - 2 = 2

i = 3 3 ≤ 4

i = 3 j = 1 = space

j ≤ 4

i = 3 j = 2 = 1

2 ≤ 4

i = 3 j = 3 = 2

3 ≤ 4

i = 3 j = 4 = 3

4 ≤ 4

~~i = 3 j = 5~~

5 ≤ 5 ✗

n - i = space

4 - 3 = 1 space

(2 × i) - i)

2 × 3 - 3)

6 - 3 = 3

i = 4 4 ≤ 4

i = 4 j = 1 = 1

j ≤ 4

i = 4 j = 2 = 2

2 ≤ 4

i = 4 j = 3 = 3

3 ≤ 4

i = 4 j = 4 → 4

4 ≤ 4

j = 5 5 ≤ 4

n - i = spa

4 - 4 = 0 space

(2 × i) - i

2 × 4 - 4

8 - 4 = 4

n - 1 = 4 - 1 = 3

i = 1 j = 1 = spce

i = 1 j = 2 = 1

i = 1 j = 3 = 2

i = 1 j = 4 3

~~n - i = 1~~

~~4 - 1 = 3~~

$\boxed{n - i \quad 4 - 3 = 1}$

i = 2 j = 1 = space

i = 2 j = 2 = space

i = 2 j = 3 = 1

i = 2 j = 4 = 2

$\boxed{n - i, 4 - 2 = 2}$

i = 3 j = 1 = space

i = 3 j = 2 = space

i = 3 j = 3 = space

i = 3 j = 4 = 1

$\boxed{n - i \quad 4 - 3 = 1}$

$\boxed{(2 \times 3) - 3 \quad 6 - 3 = 3}$

$\boxed{2 \times 2 - 2 \quad 4 - 2 = 2}$

$\boxed{2 \times 1 - 1 \quad 2 - 1 \quad 1}$

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Q.2] Numbers Butterfly n=4

```
import java.util.*;
Class Demo {
public static void main (string args[])
 Scanner scan = new Scanner (System.
 int n = ~~no~~ scan. nextInt();
 for ( int i=1 ; i ≤ n ; i++) {
   for (int j = 1 ; j ≤ i ; j++ {
      sop ( j+" ");  }
   for (int j = 1 ; j ≤ 2(n-i) ; j++){
      sop (" _ ");
   } for (int j = i ; j >= 1 ; j --){
      sop (j+" _ ");
   sopln();
 }
}

for (int i = n ; i >= 1 ; i --) {
   for (int j = 1 ; j <= i ; j++) {
      sop (j + " ");
 }
 for (int j = 1 ; j ≤ (2(n-i)) ; j++
      sop (" ");
 }
 for (int j = i ; j ≥ 1 ; j --{
   } sop (j+" _ ");
 } } sopln();
```

$i=1 \quad j \le i$

output

1	1
12	21
123	321
1234	4321
1234	432
123	32
12	2
1	1

for int i=1

~~j=1 sop(1)~~

j=2

upper part

i=1 j=1 → 1

i=2 j=1 j=2 → 1 2

i=3 j=1 j=2 j=3 ⇒ 1 2 3

i=4 j=1 j=2 j=3 ⇒ 1 2 3 4

$\boxed{i=1}$ $i \le 6$

j=1 $j \le (2*n)-i) = (2 \times (4-1)) \; 2 \times 3$

$= \cancel{2 \times 2 = 8 = 2} \; 6$

j=2

j=3 ------ = 6 ⇒ spac

j=4 $i \le 4$

$\boxed{i=2}$ ~~j=1 j≤2~~ $j \le 2(n-i) = 2 \times (4-2) = 2 \times 2 = 4$

j=2 ---- = 4 spac

j=3

j=4 $i \le 2$ $= 2 \times (4-3) = 2 \times 1 = 2$

$\boxed{i=3}$ j=1 = 2 space

j=2

$i \le 0$

$\boxed{i=4}$ j = 0

lower part

for $\boxed{i=n}$ $i \ge 1$ j---

i=4 $j \le i$

j=1 = 1

j=2 = 2

j=3 = 3

j=4 = 4

i=3 $j \le 3$

j=1 = 1

j=2 = 2

j=3 = 3

i=2 $j \le 2$

j=1 = 1

j=2 = 2

$\boxed{i=1}$ = j

j=1 $j \ge 1$ = 1

$\boxed{i=2}$ $2 \le 1$

j=2, j=1 = 2 1

i=3

$\boxed{i=3}$ $3 \le 1$

j=3, j=2, j=1 321

$\boxed{i=4}$ $4 \le 1$

j=4, j=3 j=2 j=1 4321

i=1 $j \le 1$ space $(2(n-i)$

j=1 = 1

$\boxed{i=1}$

j=1 $(2(4-(4))$ space

$= 2 \times 0 = 0$

j=2 $2(4-3)$

$2 \times 1 = 2$ space

j=3 $2(4-2)$

$2 \times 2 = 4$

j=4 $2(4-1)$

$2 \times 3 = 6$

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Q.3] Cross (x) Number Pattern
for n = 5 ;

```
import
class
pu
Scanner
int
sop
for (int i=1 ; i≤n ; i++) {
    for (int j=1 ; i≤n ; j++) {
      if (i==j || j==n-i+1)
      sop (i+"_");
      } else {
        sop(" _");
    }
  }
  soln();
  }
}
}
```

1 1
2 2
3
4 4
5 5

<u>print j</u>

$1 \le 5$ $2 \le 5$ $3 \le 5$

int i = 1	i = 2	i = 3
i=1 j=1 = True	i=2 j=1 ⇒ F	i=3 j=1 ⇒ F
i=1 j=2 = F	i=2 j=2 ⇒ T 2	i=3 j=2 ⇒ F
i=1 j=3 = F	i=2 j=3 ⇒ F	i=3 j=3 = T
i=1 j=4 = F	i=2 j=4 ⇒ T 2	i=3 j=4 ⇒ F
i=1 j=5 = True	i=2 j=5 ⇒ F	i=3 j=5 ⇒ F

$4 \le 5$ $5 \le 5$ $i=6$ $6 \le 5$ False

i = 4	i = 5
i=4 j=1 = F	i=5 j=1 ⇒ T
=4 j=2 ⇒ T	i=5 j=2 ⇒ F
=4 j=3 = F	i=5 j=3 ⇒ F
=4 j=4 ⇒ T	i=5 j=4 ⇒ F
=4 j=5 = F	i=5 j=5 ⇒ T

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Q.6] Diamond with continuous numbers

n=4

```
import java.util.*;
class Demo{
public static void main (String args[]){
Scanner scan = new Scanner (System.in);
System.out.print ("Enter a value for n:");
int n = scan.nextInt();
for (int i=1 ; i<=n; i++){
  for (int j=1 ; j<=n-i ; j++){
     sop(" - ");
  }
  for (int j=1 ; j<=(2*i)-1 ; j++){
     sop (i+" * ");
  }
  sopln();
  }
}
for (int i=n-1; i>=1 ; j--){
  for (int j=1 ; j<= n-i ; j++ ){
     sop (" - ");
  for ( int j=1 ; j<= 2x(n-i)+1 ; j++){
     sop (j+" ");
  }
  sopln();
}
```

n=4

upper part

print j

output

1
2 3
4 5 6
7 8 9 10
11 12 13 14
14 15
16

i=1

j=1
j≤n-i
1≤4-1
1≤3

j=2
j≤4-1
1≤2

j=3
1≤4-3
1≤1

1≤0
false

i=2

j=1
j≤n-1
1≤4-2
1≤2

j=2
1≤4-2
1≤2

space

i=3

j=1
j≤n-i
1≤4-3
1≤1

space

i=1

j=1
j≤(2×i)-1
(2×1)-1
2-1

i=2

j=1
2×(2-1)
2×1-1
2×1=2

1
2 3

j=2
2×(2-1)
2×1
2

4 5 6
7 8 9 10

i=3

j=1
2×3
6-3
3

j=2

j=3

i=4

j=1
j=2
j=3
j=4

(2×4)-4
8-4
4

print j

lower part

~~i=n-1=3~~

~~j=1~~ ~~j=2~~ ~~j=3~~

~~n-i~~
~~≤4-3~~

~~i=2~~

~~j=1~~ ~~j=2~~

~~j≤n~~
j≤2

~~i=1~~
~~j=1~~

~~print j~~

~~i=1~~
j=(2×i)-1
(2×1)-1
2×1-1
2×1=1

i=3

j=1
(2×i-1)
2×(3-1)
2×2=4

print j

output

1
3
5 6
8 9 10
12 13 14
15
16

n=4

Lower

i=3

N=1 n-1

j≤n-i 4-3 = 1

1 = 3

i=2

j=1 n-i 4-2

j=2 n-i 3 2

2<1

i=1

j=1 n-i 4-1

j=2 n-i

j=3 n-i

Print j

(2*i)-1

i=1 =

j=1 = (2×n)-1°

= 2-1

1

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Q.7] Diamond with continuous numbers
n=4

```
import java.util*
class Demo {
 public static void main (String args[]){
  Scanner scan = new Scanner (System.
 sop ("Enter a number for n:");
 int n = scan.nextInt();
 for (int i=1 ; i≤n ; i++) {
   for(int j=1 ; j≤n-i ; j++) {
    sop ("_");
   for (int k=1 ; k≤i ; k++) {
     int num = i+k;
    if (num % 2 ==0) {
      sop ("1") ;
   } else {
     sop ("0") ;
   }
   sopln();
 }
}

for (int i=n-1 ; i≥1 ; i--) {
 for (int j =1 ; j≤ n-i ; j++ )
 sop ("_");
}
```

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```
for (int k = 1 ; k ≤ (2*i) - i ; i++);
~~for (int~~ int num = i+k;
        if (num % 2 == 0) {
            sop ("1");
        } else {
        sop ("0");
    }
}
~~for (int~~                    n = 4
}
}
```

upper part

output

```
1
0 1
1 0 1
0 1 0 1
1 0 1
0 1
1
```

int i = 1

j = 1 n - i (4 - 1) = 3 ---

j = 2

j = 3

i = 2

j = 1 n - i (4 - 2) = 2 ---

j = 2

i = 3 n - i (4 - 3) = 1 ---

j = 1

if (num % 2 == 0)
sop (1)
else
sop (0)

i = 1 n - i

k = 1 i

k ≤ i 1 ≤ 1 1 ≤ 1

(i + k) % 2 == 0

(1) i = 2 1

i = 2 k ≤ i

k = 1 i = 2 2 ≤ 2

(i + k) false

0

0

k = 2 i = 2

(i + k) % 2 == 0 Tr

0 1

i = 3 k ≤ i 3 ≤ 3

0 1

k = 1 i = 3

i + k = 4 = True

k 2 i = 3

(i + k) 5 % 2 == 0 False

0

0 1
1 0

k 3 i = 3

(i + k) 6 % 2 = 0

0 1
1 0 1

i = 4 4 ≤ 4

k = 1 i = 4

(1 + 4) % == 0 False

0 1
1 0

k = 2 i = 4

(2 + 6) % 2 = 0 →

0 1
1 0 1
0 1

k = 3 i = 4

(3 + 4) % 2 = 0 False

0 1
1 0 1
0 1 0

k = 4 i = 4

(4 + 4) % 2 == 0 Tr True

0 1
0 1 0
0 1 0 1

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Lower Part

n = 4 i = 3

$\boxed{i = 3}$

j = 1 n-i 4-3 = 1 —

$\boxed{i = 2}$

j = 1 n-i 4-2 = 2 — —
j = 2

$\boxed{i = 1}$

j = 1 n-i 4-1 = 3 — — —
j = 2
j = 3

```
if (i+j % 2 == 0)
    sop (1)
else
    sop(0)
```

Prime space $k \leq 3$

n = 4 $\boxed{i = 3}$

k = 1, i = 3

i + k = 2 % True $\boxed{1}$

k = 2, i = 3

(2+3) % False $\boxed{10}$

k = 3 i = 3

(3+3) % True $\boxed{101}$

(2 * i) + i

2 × 3 + i

6 + 1

k < 2 × 3 + 1

k < 4 + 1

k < 5

k < 2 × 1 + 1

$\boxed{i = 2}$ $k \leq 2(2 * i) - i)$

= 2

k = 1, i = 2

(i + k) % 2 = 0 False $\boxed{10}$

k = 2 i = 2

(2+2) % 2 == 0 Tr $\boxed{\begin{matrix} 101 \\ 01 \end{matrix}}$

$\boxed{i = 1}$ $k \leq 1 (2 * i) - i)$

= 1

k = 1, i = 1

(i+k) % 2 == 0 True

$\boxed{\begin{matrix} 101 \\ 01 \\ 1 \end{matrix}}$